

Autonomous Control Performance

Introduction:

The husky vehicle is considered an under actuated system. Under actuated systems are those systems where the number of control inputs is less than the number of the degrees of freedom. Controlling such systems is a difficult task because there are not enough inputs to control all the degrees of freedom. Two different controllers have been used on the husky robot to overcome this problem. The results of the two controller, Point-to-Point Controller and Lyapunov based controller have been presented and compared

Point-to-Point (P2P) controller:

It is a state feedback type controller. A transformation has been made from the coordinates of the system x, y, θ to polar coordinates ρ, β, α to facilitate the construction of the controller. The control law used for the linear velocity $v = k_\rho * \rho$ and for the angular velocity $\omega = k_\alpha * \alpha + k_\beta * \beta$

Lyapunov Controller :

It is an energy based approach. A selected function is used to represent the energy of the system and its derivative represents the decaying of that energy till it reaches zero at the desired position. The control law for the linear velocity $v = v_d * \cos(\tilde{\theta}_r) + k_x * \tilde{x}_r$ and for the angular velocity $\omega = v_d * \tilde{y}_r + \omega_d + k_\theta * \sin(\tilde{\theta}_r)$

Test Cases and Parameters :

The following table represents different cases for the controllers. The table shows the initial conditions as well as the desired value for x, y and θ .

Case Num	x_0	y_0	θ_0	x_{desire}	y_{desire}	θ_{desire}
1	2	0	0	10	10	90
2	2	0	180	-3	5	90
3	2	-3	0	1	3	180
4	6	-8	0	-6	8	0
5	-6	9	0	4	3	180
6	-3	5	90	-6	8	90
7	-5	-3	180	5	3	90
8	-3	-5	180	-6	8	180

The following table shows the values of the parameters used in both controllers :

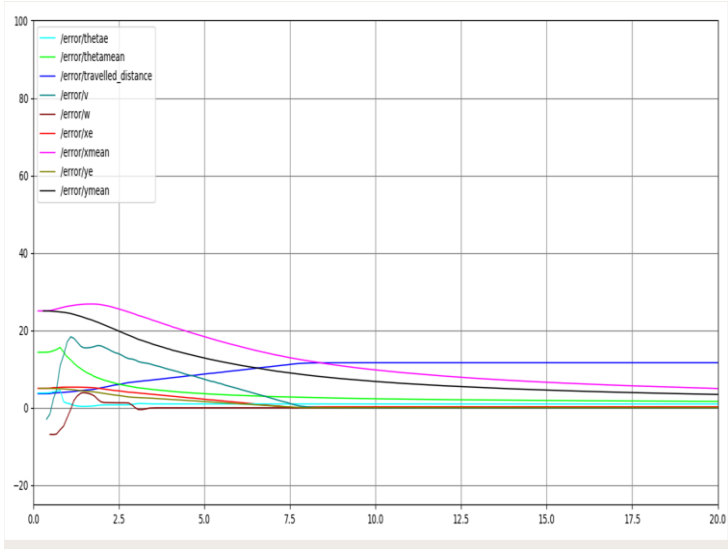
Controller Type	k_ρ	k_α	k_β	Controller Type	k_x	k_θ	v_d	ω_d
P2P	0.5	1.5	-0.6	Lyapunov	2.5	0.5	1	0.5

Figure 1: Parameters' Values

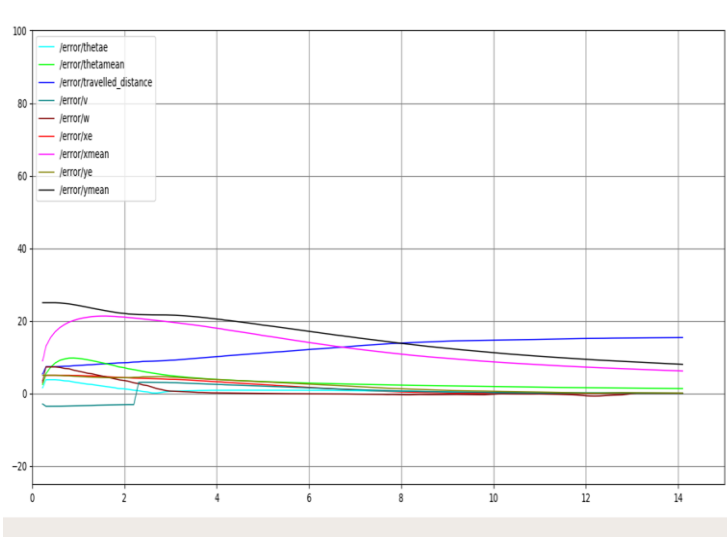
Results and Figures

In this section, the results of both controllers will be presented and compared.

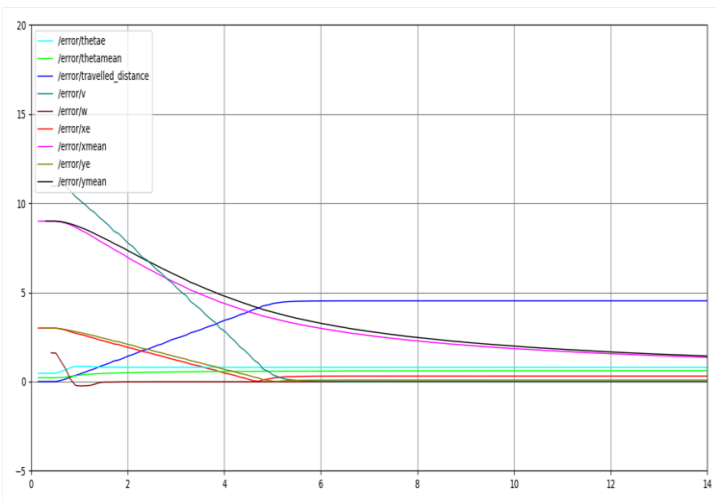
Test case 2 Lyapunov:



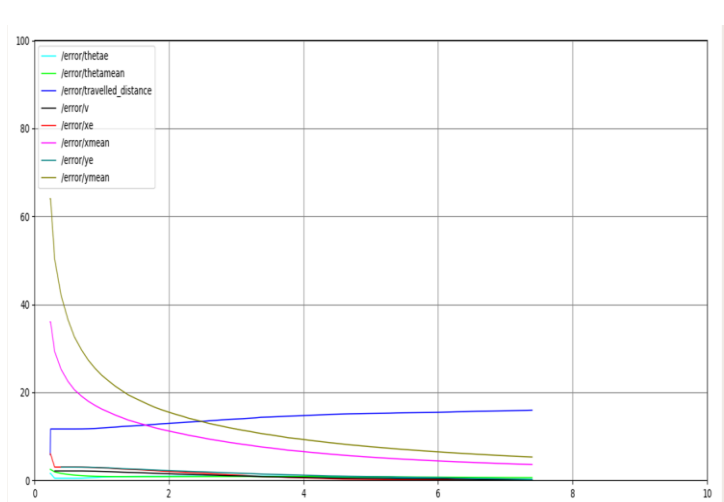
Test case 2 point to point:



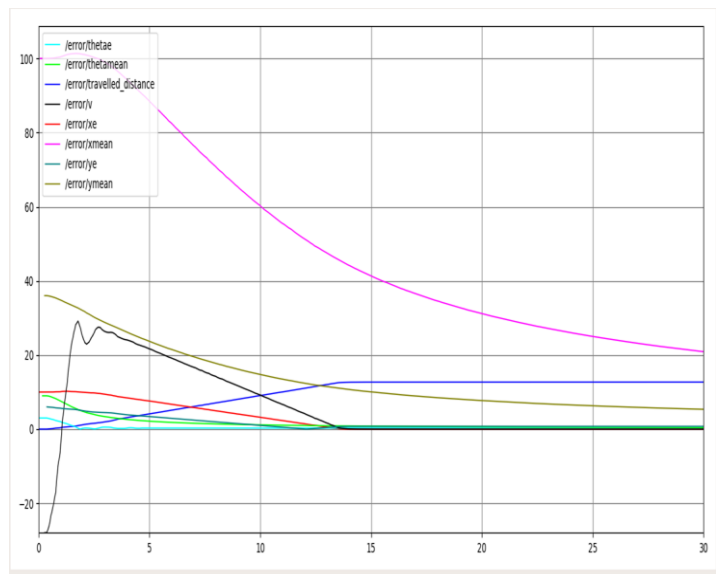
Test case 4 Lyapunov:



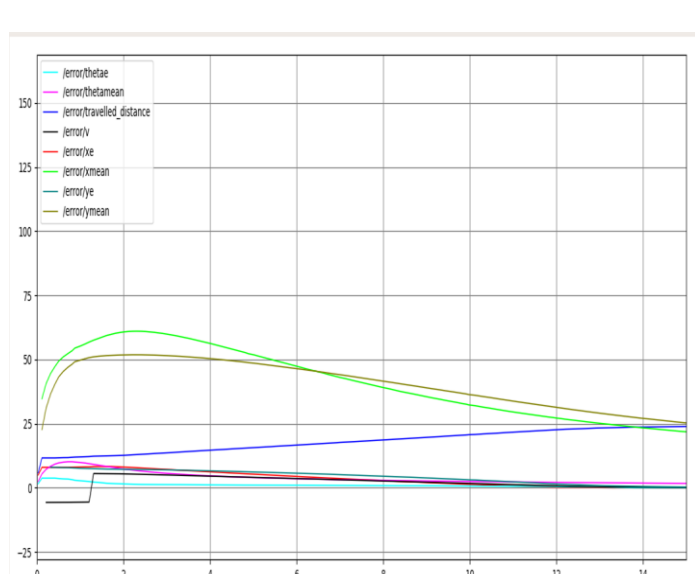
Test case 4 point to point:



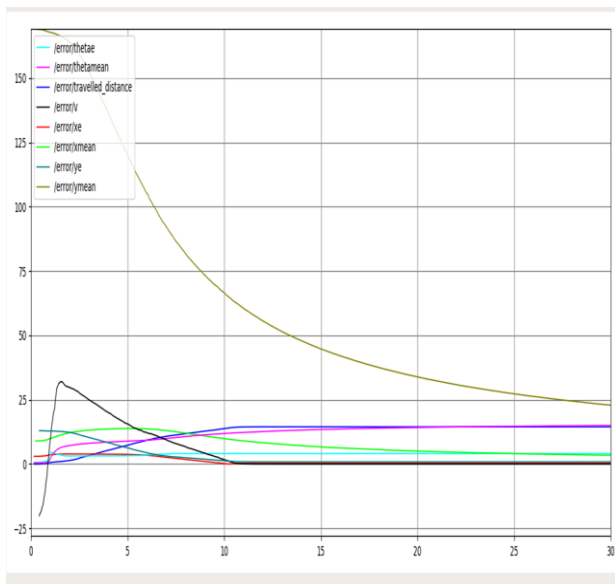
Test case 6 Lyapunov:



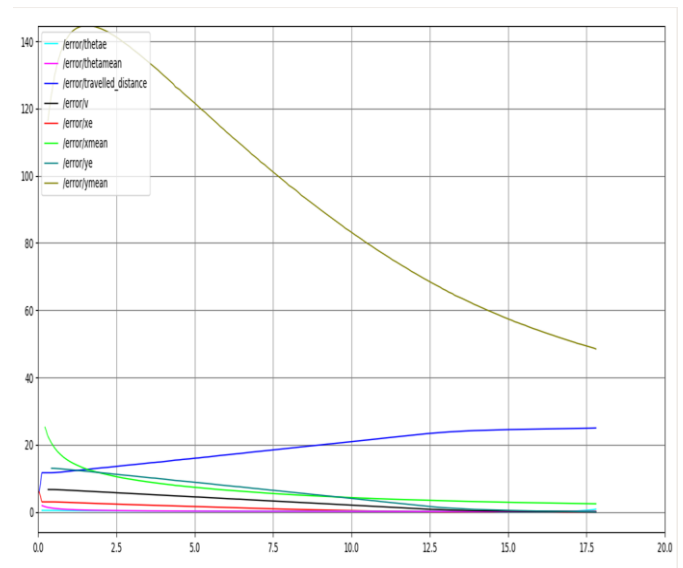
Test case 6 point to point:



Test case 8 Lyapunov:



Test case 8 point to point:



Discussion:

The results show that both of the controllers were able to reach the desired values, but with steady state errors. Lyapunov controller showed a more stable behavior than the point to point controller which was not robust enough in some positions and causes oscillations in the error values and the controller effort. Furthermore, Lyapunov could have better performance if there was a desired trajectory instead of a fixed desired position as this would help to get the inverse kinematics of the robot which results in getting the desired linear and angular velocity required to reach the desired position with much less error than the one happened during the simulation.

Map Design using Build Editor:

